Math Check #5 Study Sheet: Ms. Womack's Model

Part 1 · Imagine you have two stocks, A and B. How much money goes into each?

Decision variables

$x_A, x_B =$ the proportion of the money invested in each stock

Example: x_A = 0.25 means 25% of the money goes into stock A.

Objective function

maximize expected return $= \bar{r}_A x_A + \bar{r}_B x_B$

$\bar{r}_A$ is stock A's average monthly return. It comes from the data, so it is not a decision. Example: $\bar{r}_A$ = 0.012 means stock A gained 1.2% in an average month.
Each stock's return is weighted by the proportion invested in it. Example: with $\bar{r}_A$ = 0.012, $\bar{r}_B$ = 0.008, x_A = 0.25, x_B = 0.75, expected return = 0.012 × 0.25 + 0.008 × 0.75 = 0.009, or 0.9% a month.

Constraints

Invest all the money $x_A + x_B = 1$

Keep risk under the limit add up every cell of the weighted covariance matrix:

	A	B
A	$x_A x_A \text{Cov}(A,A)$	$x_A x_B \text{Cov}(A,B)$
B	$x_B x_A \text{Cov}(B,A)$	$x_B x_B \text{Cov}(B,B)$

$$x_A x_A \text{Cov}(A,A) + x_A x_B \text{Cov}(A,B) + x_B x_A \text{Cov}(B,A) + x_B x_B \text{Cov}(B,B) \leq \text{risk limit}$$

Blue cells: each stock with itself (its variance). The grid is symmetric: Cov(A,B) = Cov(B,A).

Stay between 0 and 1 $0 \leq x_A \leq 1, \quad 0 \leq x_B \leq 1$

Part 2 · Now add business rules, with many stocks: A, B, C, and so on

We write each rule for stock A. The model repeats it for every other stock. The objective function and the constraints from Part 1 stay the same, just with more stocks.

New decision variable

$y_A = 1$ if we buy stock A, $\quad y_A = 0$ if we do not $\quad$ (a binary decision variable)

x_A is how much we invest in stock A. y_A is whether we buy it at all.

Linking constraints

If we buy stock A, at least 2% $x_A \geq 0.02\, y_A$

Never more than 10% in stock A $x_A \leq 0.10\, y_A$

They link how much (x) to whether (y). y_A can only be 0 or 1, so plug in each one:

If y_A = 0 (skip stock A): the rules become $x_A \geq 0.02 \times 0 = 0$ and $x_A \leq 0.10 \times 0 = 0$. The only number that is both ≥ 0 and ≤ 0 is 0, so **x_A = 0**: no money goes into stock A.

If y_A = 1 (buy stock A): the rules become $x_A \geq 0.02 \times 1 = 0.02$ and $x_A \leq 0.10 \times 1 = 0.10$, so **stock A gets between 2% and 10%** of the money.

Why we need y at all: the rule is "either nothing, or at least 2%." Writing just $x_A \geq 0.02$ would force us to buy stock A. The binary y_A lets the model choose which case it wants.

Pick at least 2 Tech stocks

Suppose A, B, and C are our three Tech stocks: $y_A + y_B + y_C \geq 2$

Each y is 1 if we buy that stock and 0 if we do not, so **adding the y's counts how many Tech stocks we bought**.
Buy A and C, skip B: 1 + 0 + 1 = 2, so the rule is met. Buy only A: 1 + 0 + 0 = 1, so the rule is broken.
This is a project-selection constraint. It works for any sector and any number: change the stocks in the sum and the 2.

Why Part 2 needs more than two stocks: with a 10% cap you must hold at least **10** stocks (10 × 10% = 100%). With only stocks A and B you could invest at most 20%, so $x_A + x_B = 1$ would be impossible (infeasible). Binary variables + nonlinear risk means we solve with Bonmin, not ipopt.